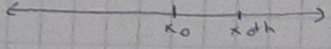
Nümerik Diferansiyelleme

iki nokta formülü

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$



$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h}$$

$f \in C^2[a,b]$ bir fonk $x_0 \in (a,b)$ bir nokta

$\boxed{x_1 = x_0 + h} \in [a,b]$ olacak şekilde bir h alalım.

x	$f(x)$
x_0	$f(x_0)$
x_1	$f(x_1)$

$$\underbrace{L_{1,0}(x)}_{\frac{x-x_1}{x_0-x_1}} f(x_0) + \underbrace{L_{1,1}(x)}_{\frac{x-x_0}{x_1-x_0}} f(x_1) = P_1(x)$$

$$P_1(x) = \frac{x-x_0-h}{-h} f(x_0) + \frac{x-x_0}{h} f(x_0+h),$$

$$f(x) = P_1(x) + f''(\xi)(x-x_0)(x-x_0-h)$$

$$P_1'(x) = \frac{-f(x_0)}{h} + \frac{f(x_0+h)}{h} \approx f'(x_0)$$

$$f'(x) = \frac{f(x_0+h) - f(x_0)}{h} + \frac{2(x-x_0)-h}{2} f''(\xi) + \left(\frac{f''(\xi)}{2}\right)'_x (x-x_0)(x-\cancel{x_0}-\cancel{h})$$

$$f'(x_0) = P_1'(x_0) \underbrace{-\frac{h}{2} f''(\xi)}_{\text{hata terimi}} \qquad x_0 \leq \xi \leq x_0+h$$

Not 1 $f(x)$'in x_0 noktasından taylor formülü

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2!}(x-x_0)^2, \quad \xi \in (x_0, x)$$

$x = x_0 + h$ yazalım

$$f(x_0+h) = f(x_0) + f'(x_0)(x_0+h-x_0) + \frac{f''(\xi)}{2}(x_0+h-x_0)^2$$

$$= f(x_0) + f'(x_0)\,h + h^2 \frac{f''(\xi)}{2} \qquad \xi \in (x_0, x_0+h)$$

$$f'(x) = \frac{f(x_0+h) - f(x_0)}{h} - h\frac{f''(\xi)}{2} \qquad \xi \in (x_0, x_0+h)$$

Örnek

	x	$f(x)$	$f'(x)$	$f'(x)$ geri
ileri ← 0,1	0,5	0,45	0,8	
ileri / geri ←	0,6	0,56	1	0,8
geri ← 0,1	0,7	0,66		1

geri fark tüm ileri fark formüllerinde
$h \Rightarrow -h$ konularak bulunur

Örnek 2

$f(x) = \ln x$, $x_0 = 1,8$ noktasında 1. mertebeden türevi

$h = 0,01$ olacak biçimde tahmini hesaplayıp hatasını

değerlendiriniz

$$f'(1,8) = \frac{f(1,8+0,01) - f(1,8)}{0,01} = \frac{\ln(1,81) - \ln(1,8)}{0,01}$$

$$= 0,5540$$

$$R(1,8) = -\frac{h}{2} f''(\xi) \quad \xi \in (1,8 , 1,81)$$

$f(x) = \ln x$, $f'(x) = \frac{1}{x}$, $f''(x) = -\frac{1}{x^2}$

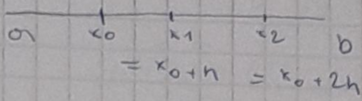
$$= \left|\frac{h^2}{2} \cdot \frac{1}{\xi^2}\right| = \left|\frac{0,01}{2} \cdot \frac{1}{\xi^2}\right|$$

$$\leq \frac{0,01}{2} \cdot \frac{1}{(1,8)^2} \leq 1,5 \cdot 10^{-3}$$

$$0,5525 \leq f'(1,8) \leq 0,5555$$

$$\frac{1}{1,8} \approx 0,5555$$

$f \in C^3[a,b]$ ve $x_0 \in (a,b)$, $h \neq 0$ adımı ile $x_1 = x_0 + h$, $x_2 = x_0 + 2h \in [a,b]$ tanımlansın



$$P_2(x) = L_{2,0}(x) f(x_0) + L_{2,1}(x) f(x_1) + L_{2,2}(x) f(x_2)$$

$$\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$$

$$\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \qquad \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$f(x) = P_2(x) + \frac{f'''(\xi)}{3!}(x-x_0)(x-x_1)(x-x_2)$$

$$f'(x) = P_2'(x) + \left(\frac{f'''(\xi)}{3!}\right)'_x (x-x_0)(x-x_0-h)(x-x_0-2h)$$
$$+ \frac{f'''(\xi)}{6}\Big[(x-x_0-h)(x-x_0-2h) + (x-x_0)(x-x_0-2h) + (x-x_0)(x-x_0-h)\Big]$$

$$P_2'(x) = \frac{2x-(2x_0+3h)}{2h^2} f(x_0) + \frac{2x-(2x_0+2h)}{-h^2} f(x_0+h) + \frac{2x-(2x_0+h)}{2h^2} f(x_0+2h)$$

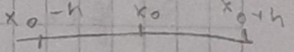
$$f'(x_0) = P_2'(x_0) + \frac{f'''(\xi)}{6} 2 \cdot h^2 \qquad x_0 < \xi < x_0 + 2h$$

3 nokta uç nokta formülü

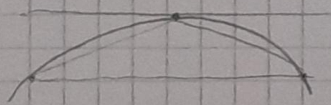
$$P_2'(x_0) = \frac{1}{2h}\Big[-3f(x_0) + 4f(x_0+h) - f(x_0+2h)\Big] \approx f'(x_0)$$

Türevde $x = x_1 = x_0 + h$

$$f'(x_0+h) = P_2'(x_0+h) + \frac{f'''(\xi)}{6}(-h^2)$$
$$P_2'(x_0+h) = \frac{1}{2h}\Big[-f(x_0) + f(x_0+2h)\Big]$$



$$f'(x_0) = \frac{1}{2h}\Big[f(x_0+h) - f(x_0-h)\Big] - \frac{h^2}{6} f'''(\xi)$$
$$x_0 - h < \xi < x_0 + h$$



Örnek 3

$f'(x) = 2,0$ daki değeri? tahmin ediniz

x	$f(x)$	$f'(x)$
1,8	13,5	
2,0	12,3	7
2,1	12,7	6,5
2,2	13,7	

3 nokta midpoint

3 nokta uç nokta

2. mertebeden türev için nümerik diferansiyelleme formülü

$f \in C^4[a,b]$ ve $x_0 \in (a,b)$ verilmiş olsun. $x = x_0$ noktasındaki

taylor formülüne göre

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \frac{f^{(4)}(\xi)}{4!}(x-x_0)^4$$

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2!}h^2 + \frac{f'''(x_0)}{3!}h^3 + \frac{f^{(4)}(\xi)}{4!}h^4$$

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2!}h^2 - \frac{f'''(x_0)}{3!}h^3 + \frac{f^{(4)}(\xi_2)}{4!}h^4$$

ikisini toplayalım

$$f(x_0+h) - f(x_0-h) = f'(x_0)2h + \frac{f'''(\xi)2h^3}{3!}$$

(midpointa 2h'a bölersek benzer aynı)

$$f(x_0+h) + f(x_0-h) = 2f(x_0) + \frac{2f''(x_0)}{2!}h^2 + \frac{f^{(4)}(\xi_1)}{4!}h^4 + \frac{f^{(4)}(\xi_2)}{4!}h^4$$

$\leftrightarrow$

$$\to \frac{f^{(4)}(\xi_1) + f^{(4)}(\xi_2)}{2} = f^{(4)}(\xi)$$

$$\to f''(x_0) = \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2} - \underbrace{\frac{h^2}{12}f^{(4)}(\xi)}_{\text{hata}}$$

$f'(x) = (1x)e^x \Rightarrow f''(x) = (2+x)e^x \Rightarrow f'''(x) = (3+x)e^x \Rightarrow f^{(4)}(x) = (4+x)e^x$

Örnek

$f(x) = xe^x$

$f(x) = xe^x \longrightarrow f''(2)$'yi tahmin edin ve hatasını belirtin

$h_1 = 0,1$ $h_2 = 0,2$

	x	$f(x)$	$f''(x)$ h_1	h_2
$x_0 - h_2$	1,8	10,89		
$x_0 - h_1$	1,9	12,70		
x_0	2,0	14,78	29	29,5
$x_0 + h_1$	2,1	17,15		
$x_0 + h_2$	2,2	19,85		

$$f''(x_0) = \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2}$$

$$\text{hata} = -\frac{h^2}{12} f^{(4)}(\xi)$$

$$R(2,0) = -\frac{h^2}{12} f^{(4)}(\xi) \qquad 1,8 < \xi < 2,2$$

$$|R(2,0)| \leq \left| \frac{(0,2)^2}{12} (4+\xi) e^{\xi} \right| \leq \frac{1}{30} (4+2,2) e^{2,2}$$

$$\approx 0,1865$$

$f''(2) = 29,5562$

uygulama

1) $f(x) = x\ln x$ fonksiyonu ve $x_0 = 1$, $x_1 = e$ düğüm noktaları verilsin

a) Hermite polinomunu belirleyiniz

b) Bu polinomu kullanarak $f(2)$'nin tahmini değerini bulunuz ve hata değerlendiriniz

a) $n = 1$ $\quad 2n+1 = 3$. dereceden hermite polinomu

$f(x) = x\ln x \Rightarrow f'(x) = \ln x + \frac{1}{x}x = \ln x + 1$

$f(x_0) = f(1) = 0$, $f(x_1) = f(e) = e\ln e = e$

$f'(x_0) = f'(1) = 1$

$f'(x_1) = f'(e) = 2$

$H_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

$H_3'(x) = a_1 + 2a_2 x + 3a_3 x^2$

$H_3(x_0) = H_3(2) = a_0 + a_1 + a_2 + a_3 = f(1) = 0$

$H_3(x_1) = H_3(e) = a_0 + a_1 e + a_2 e^2 + a_3 e^3 = e$

$H_3'(x_0) = H_3'(1) = a_1 + 2a_2 + 3a_3 = f'(x_0) = 1$

$H_3'(x_1) = H_3'(e) = a_1 + 2a_2 e + 3a_3 e^2 = f'(e) = 2$

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e & e^2 & e^3 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2e & 3e^2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 0 \\ e \\ 1 \\ 2 \end{bmatrix}}_{b}$$

$a_0 = -0,3103353$ $\qquad a_2 = 0,002054$

$a_1 = -0,0345194$ $\qquad a_3 = -0,033325$

0325+

b) $f(x) = \underbrace{H_3(x)}_{\text{hermite polinomu}} + \underbrace{R(x)}_{\text{hata terimi}}$

$R(x) = \frac{f^{(4)}(\xi)}{4!}(x-x_0)^2(x-x_1)^2$

$f(2) \approx H_3(2) = 1,3785834$

$R(x) = \frac{f^{(4)}(x)}{24}(x-1)^2(x-e)^2 \quad \xi \in (1,e)$

$f(x) = x \ln x$

$f'(x) = \ln x + 1$

$f''(x) = \frac{1}{x}$

$f'''(x) = -\frac{1}{x^2}$

$f^{(4)}(x) = \frac{2}{x^3}$

$R(x) = \frac{2}{24} \cdot \frac{1}{\xi^3}(x-1)^2(x-e)^2$

$|R(2)| = \frac{1}{12} \cdot \frac{1}{\xi^3}(2-e)^2$

$|f(2)| \leq \frac{1}{12}(2-e)^2$

2) $[0,2]$ aralığında doğal sınır koşullarına sahip kübik spline

$S(x) = \begin{cases} S_0(x) = 1 + 2x - x^3, & 0 \leq x \leq 1 \\ S_1(x) = 2 + b(x-1) + c(x-1)^2 + d(x-1)^3, & 1 \leq x \leq 2 \end{cases}$

b, c ve d yi bulunuz

$S_0(0) = 1$

$\boxed{S_0(1) = 2}$

$\boxed{S_1(1) = 2}$

doğal

$S_0'(1) = S_1'(1)$

$2 - 3x^2 \big|_{x=1} = b + 2c(x-1) + 3d(x-1$

$1 = 2 - b \Rightarrow b = -1$

$S_0''(1) = S_1''(1)$

$-6x \big|_{x=1} = 2c + 6d(x-1)$

$-6 = 2c$

$c = -3$

bağlanmış sınır koşulları

$f'(0) = f'(2)$ koşuluile $S_0'(0) = S_1'(2)$

$$2 = \underbrace{\underbrace{b}_{-1} + \underbrace{2c}_{-6}}_{-7} + 3d$$

~~$-6 = 2c$~~ $\quad d = 3$

~~$c = -3$~~

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b bilgi verilmese

$$S_0''(0) = S_1''(2)$$

$$-6x\Big|_{x=0} = 2c + 6d(x-1)\Big|_{x=2}$$

$$0 = -6 + 6d$$

Soru :

$$f'(x_0) \simeq \frac{f(x_0+h) - f(x_0)}{h}$$

ileri fark

$$f'(1) \simeq \frac{f(1,2) - f(1)}{0,2}$$

$$= 0,65$$

Geri fark

$$f'(1) \simeq \frac{f(1) - f(0,8)}{0,2}$$

$$= 0,5$$

b) Üç nokta - son nokta

$$f'(x_0) \approx \frac{1}{2h}\left[-32(x_0) + 4f(x_0+h) - f(x_0+2h)\right] = 0,6$$

Üç nokta orta nokta

$$f'(x_0) \approx \frac{1}{2h}\left[f(x_0+h) - f(x_0-h)\right] = 0,6250$$

Üç nokta - aranokta için hata

$$f'(x_0) = \frac{1}{2h}\left[f(x_0+h) - f(x_0-h)\right] - \frac{h^2}{6} f'''(\xi)$$

$O(h^2)$ $\qquad x_0 - h < \xi < x_0 + h$

2) $$f''(x_0) \approx \frac{1}{h^2}\left[f(x_0-h) - 2f(x_0) + f(x_0+h)\right]$$

$$f''(1) = \frac{1}{0,04}\left[f(0,8) - 2f(1) + f(1,2)\right] = 0,25$$